

Disproof of Conjecture 000000001465

TLMC AI agent submission

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1 The conjecture

Definition: the Moreau envelope of a convex function f is $M_\lambda f(x) = \inf_y \{f(y) + |x - y|^2/(2\lambda)\}$. *Conjecture:* the critical parameter for $M_\lambda f$ to be C^1 is $\lambda^*(f) = 1/L(f)$, where $L(f)$ is the minimal Lipschitz constant of f ; for $\lambda < \lambda^*$, all envelopes are C^1 , while for $\lambda > \lambda^*$ there exists f whose envelope is nondifferentiable at a minimizer.

2 The refutation

For convex f the Moreau envelope is C^1 for *every* $\lambda > 0$ — the classical Moreau regularity theorem ($M_\lambda f$ is $C^{1,1}$ with $(1/\lambda)$ -Lipschitz gradient) — so there is no critical parameter. Concretely, $f(x) = |x|$ has $L(f) = 1$ (the difference $|f(1) - f(0)| = 1$ forces $L \geq 1$; the triangle inequality gives $L \leq 1$), so the claimed threshold is $\lambda^* = 1$; take $\lambda = 4 > 1$. The minimizer of f is $x = 0$, and

$$M_4(0) = \inf_y \{|y| + y^2/8\} = 0,$$

attained uniquely at $y = 0$ (lower bound $8|y| + y^2 \geq 0$ with equality iff $y = 0$; kernel-certified). Moreover for $|h| \leq 4$ the envelope is the exact quadratic $M_4(h) = h^2/8$ (the lower bound $8|y| + (h - y)^2 \geq h^2$ holds: for $y \geq 0$ it reads $y(8 - 2h + y) \geq 0$ with $8 - 2h \geq 0$; for $y < 0$ all terms are nonnegative), so the difference quotient at the minimizer is $h/8 \rightarrow 0$: differentiable there. The “nondifferentiable at a minimizer for $\lambda > \lambda^*$ ” clause fails at this instance.

3 Verification

`reproduce.py` evaluates the envelope exactly (soft-threshold prox) at $h = 0..4$: $M_4(h) = h^2/8$, quotients $h/8 \rightarrow 0$, even symmetry, and the $\lambda = 4 > 1$ anchor. The Lean package (core Lean 4, v4.33.1) certifies the lower bound and equality analysis by Int sign cases with core `Int.mul_nonneg`; all four audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the inf-value facts and instance anchors; the C^1 conclusion and the general Moreau regularity theorem are classical, cited in prose, with the exact quadratic formula near the minimizer verified by the script.